

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

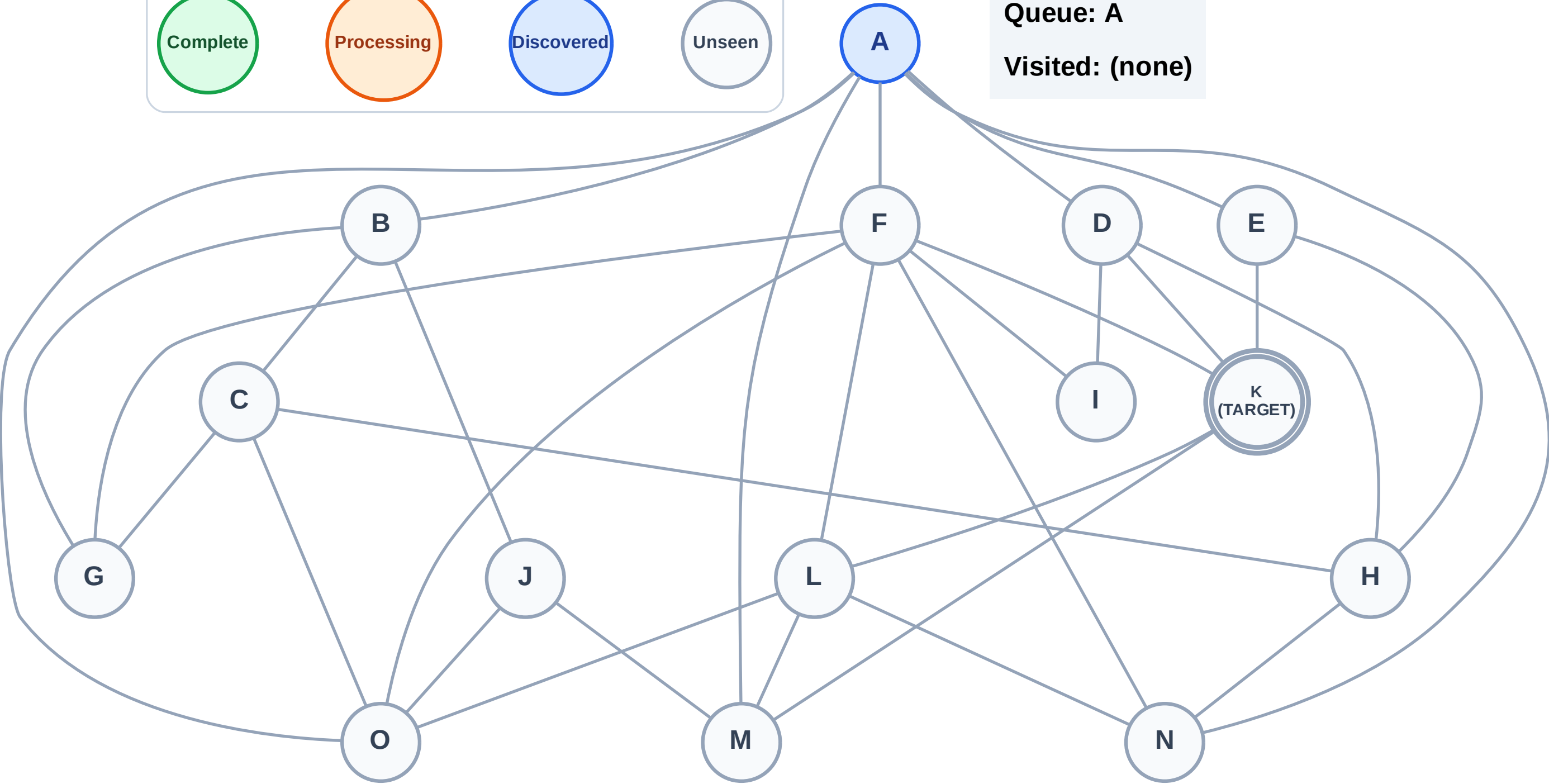
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

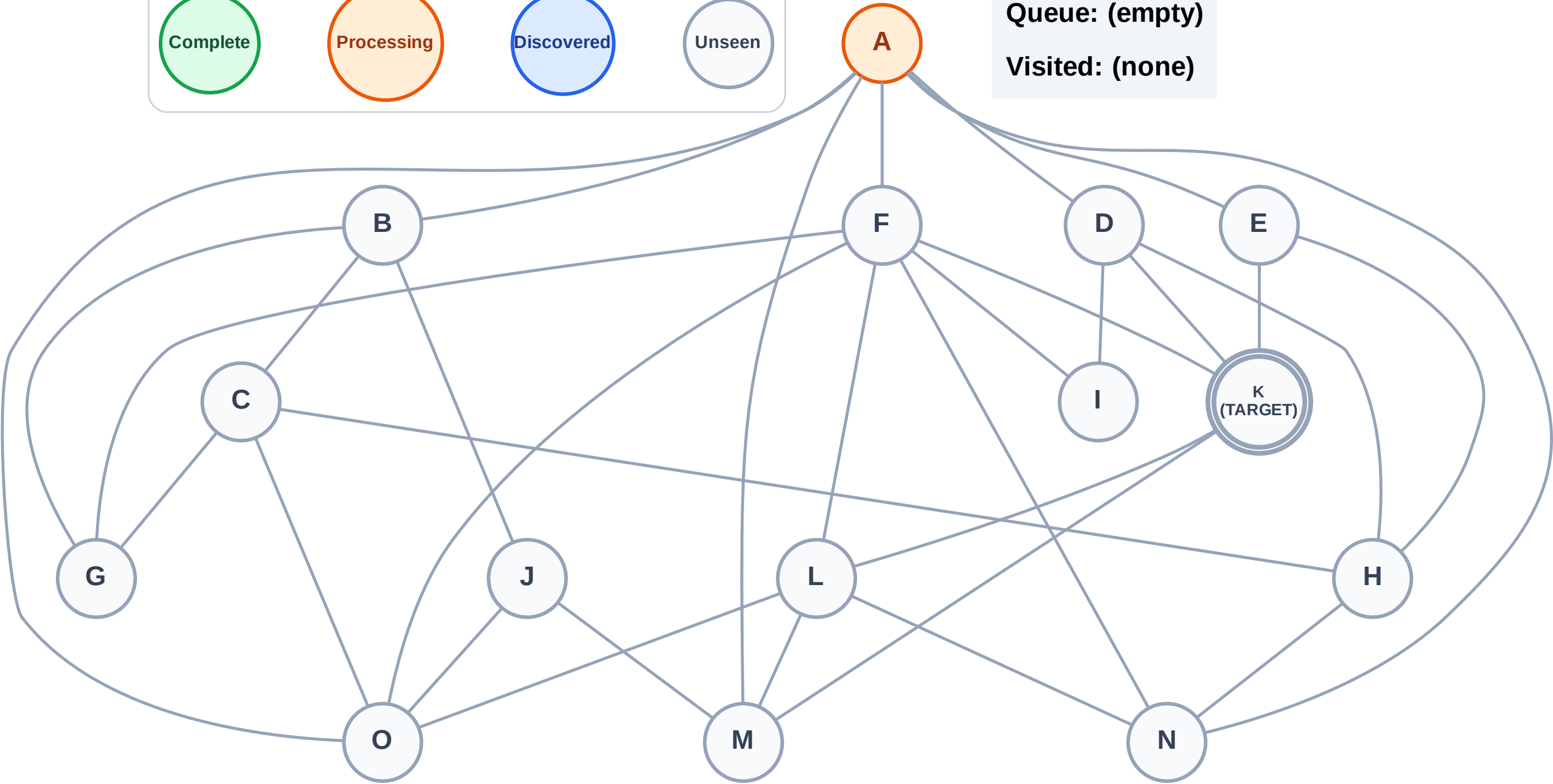
Complete

Processing

Discovered

Unseen

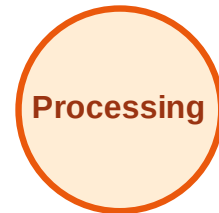
Queue: (empty)
Visited: (none)



BFS Traversal

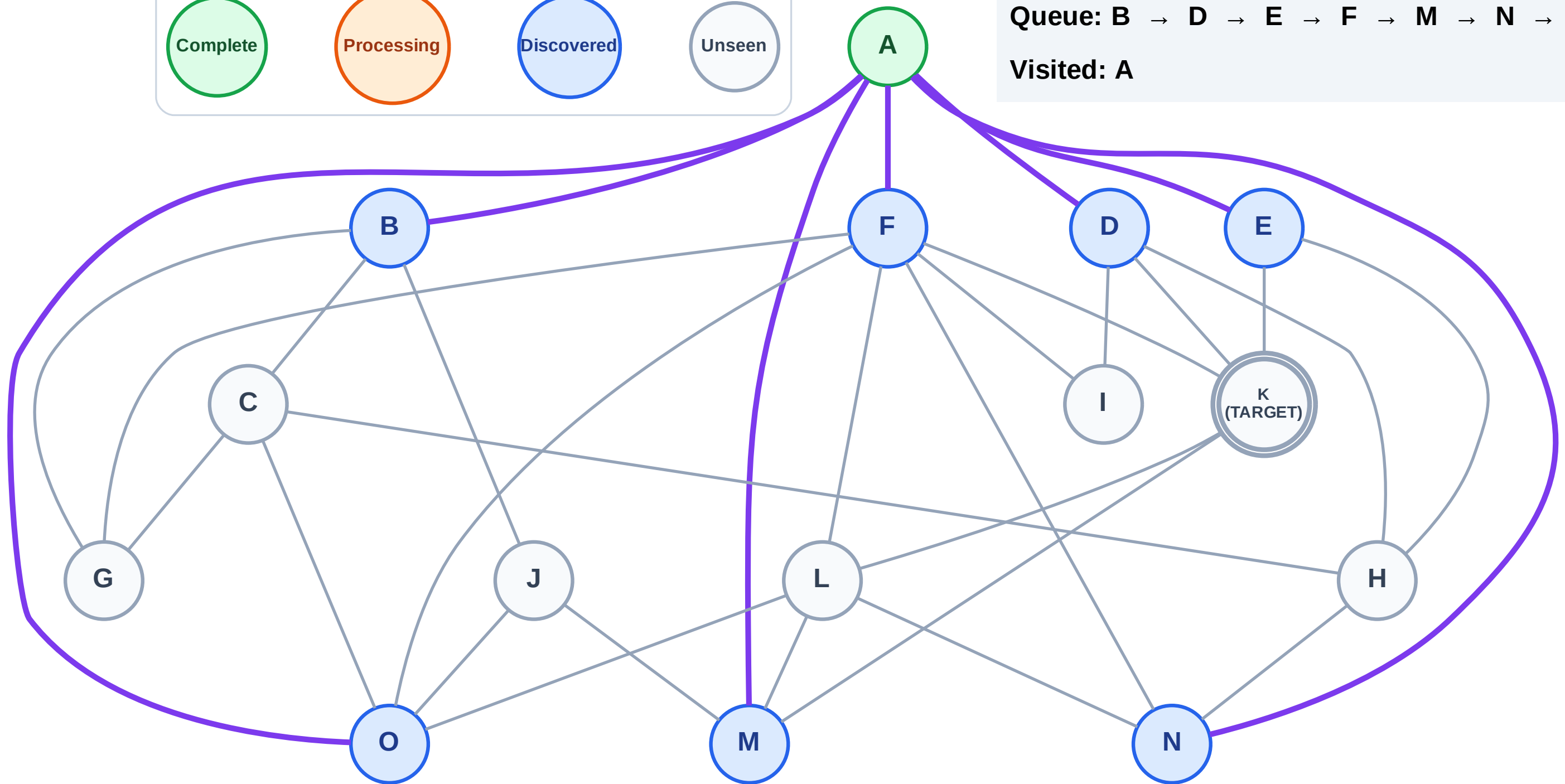
Step 3: Discover B, D, E, F, M, N, O from A

Legend



Queue: B → D → E → F → M → N → O

Visited: A



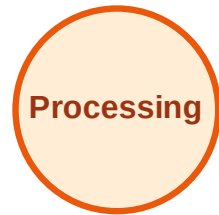
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



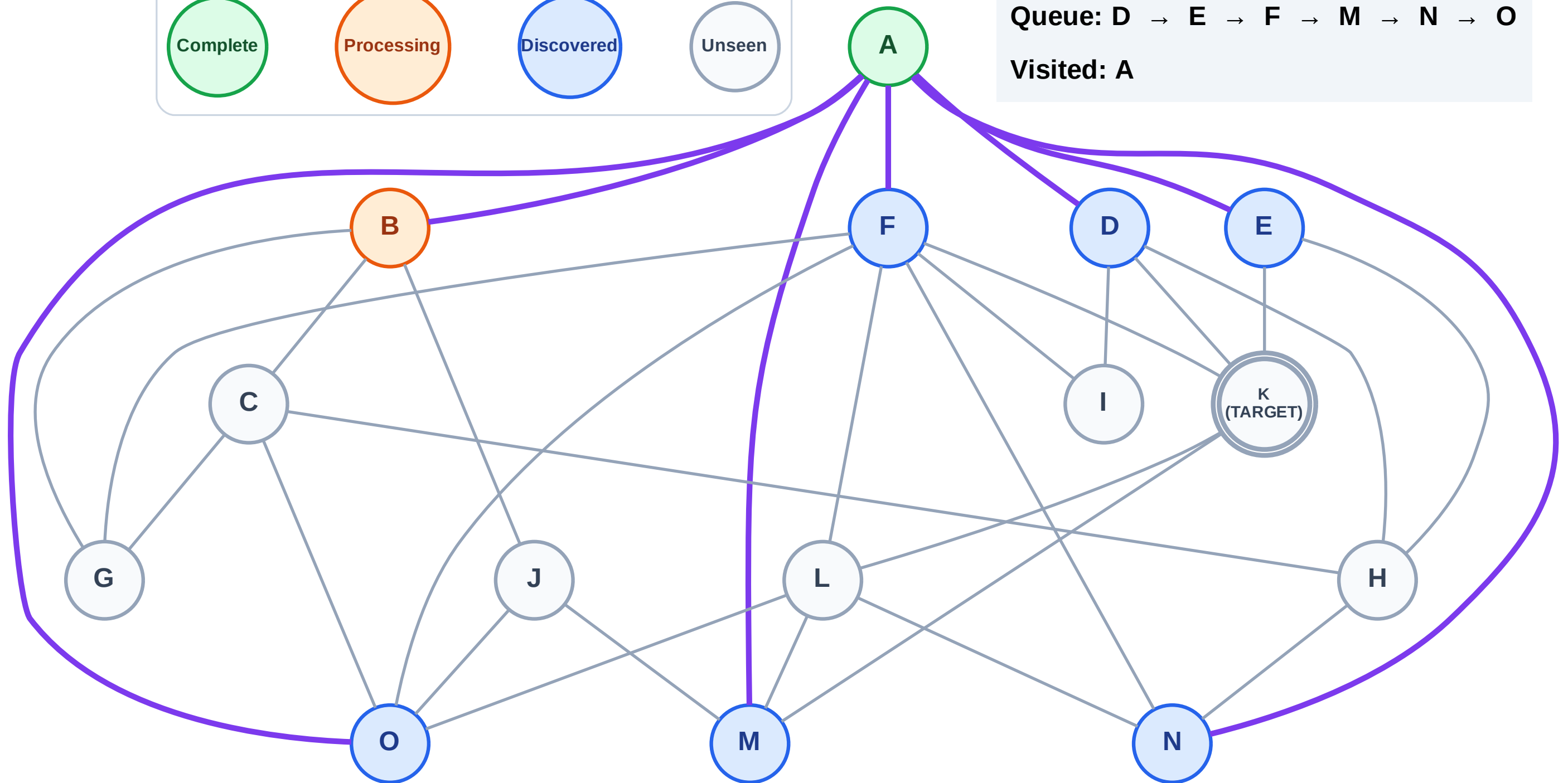
Discovered



Unseen

Queue: D → E → F → M → N → O

Visited: A



BFS Traversal

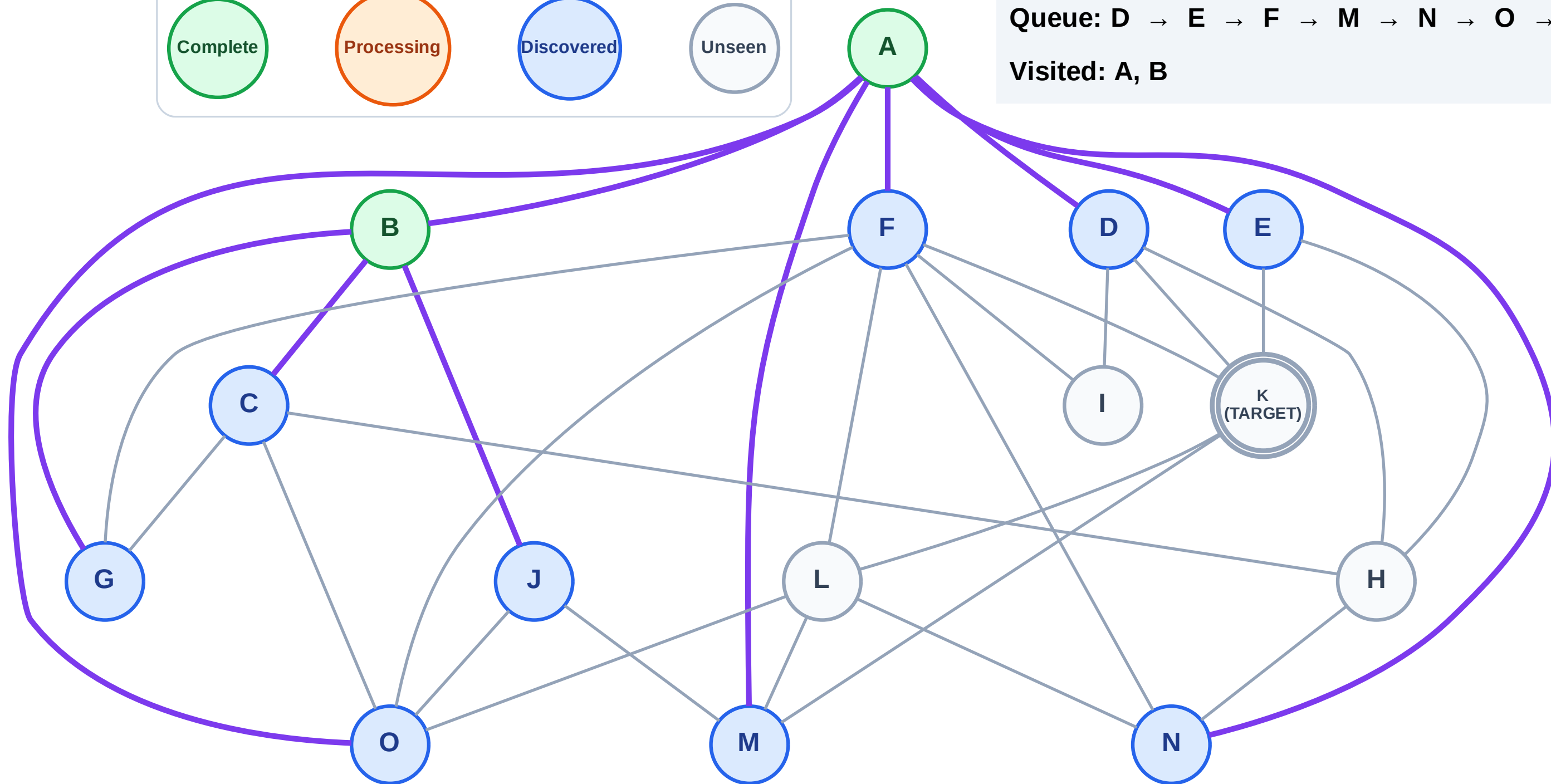
Step 5: Discover C, G, J from B

Legend



Queue: D → E → F → M → N → O → C → G → J

Visited: A, B



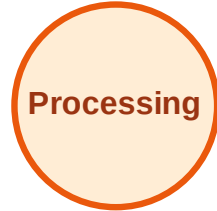
BFS Traversal

Step 6: Process node D

Legend



Complete



Processing



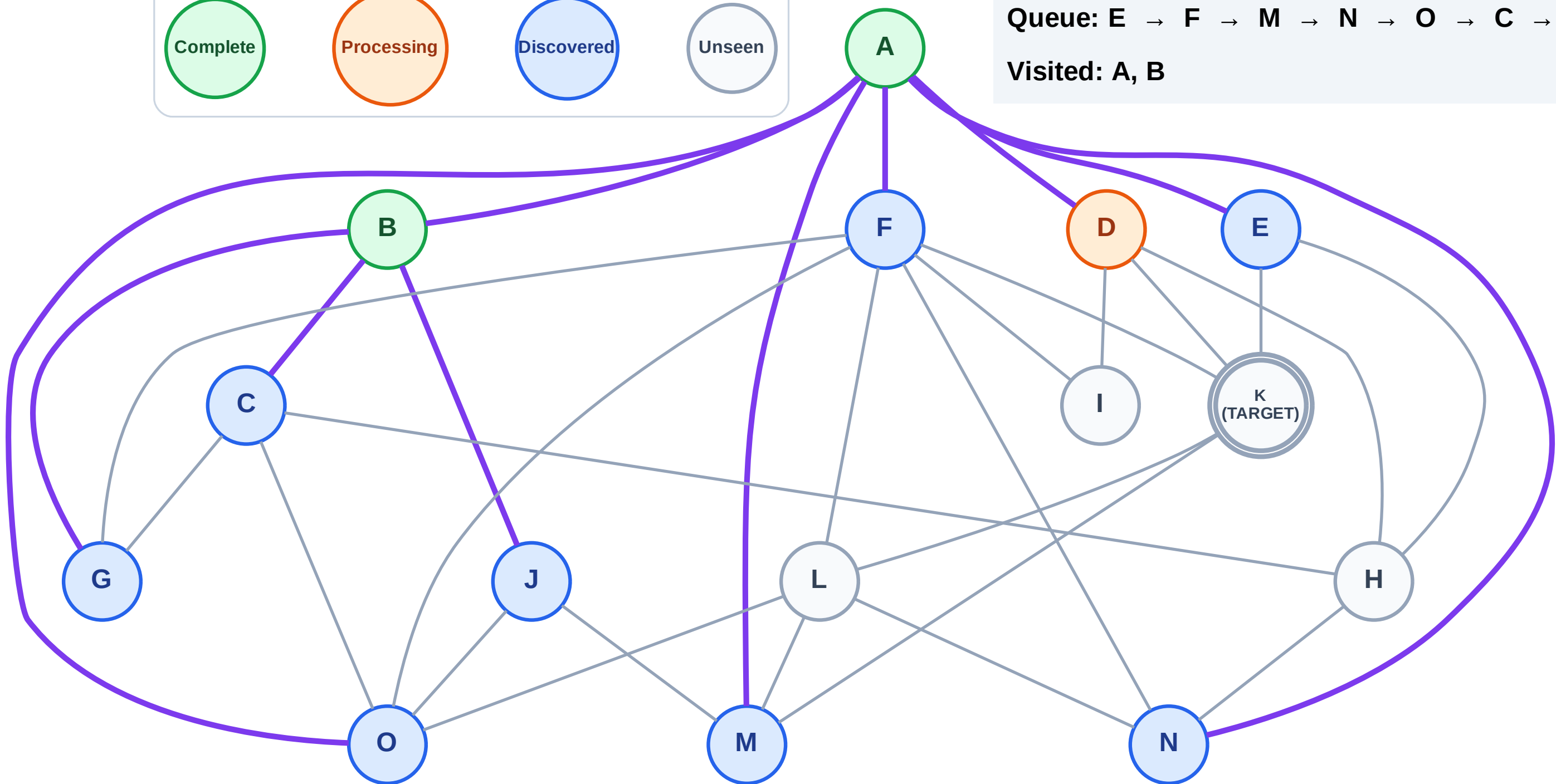
Discovered



Unseen

Queue: E → F → M → N → O → C → G → J

Visited: A, B



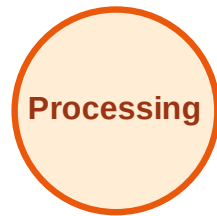
BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend



Complete



Processing



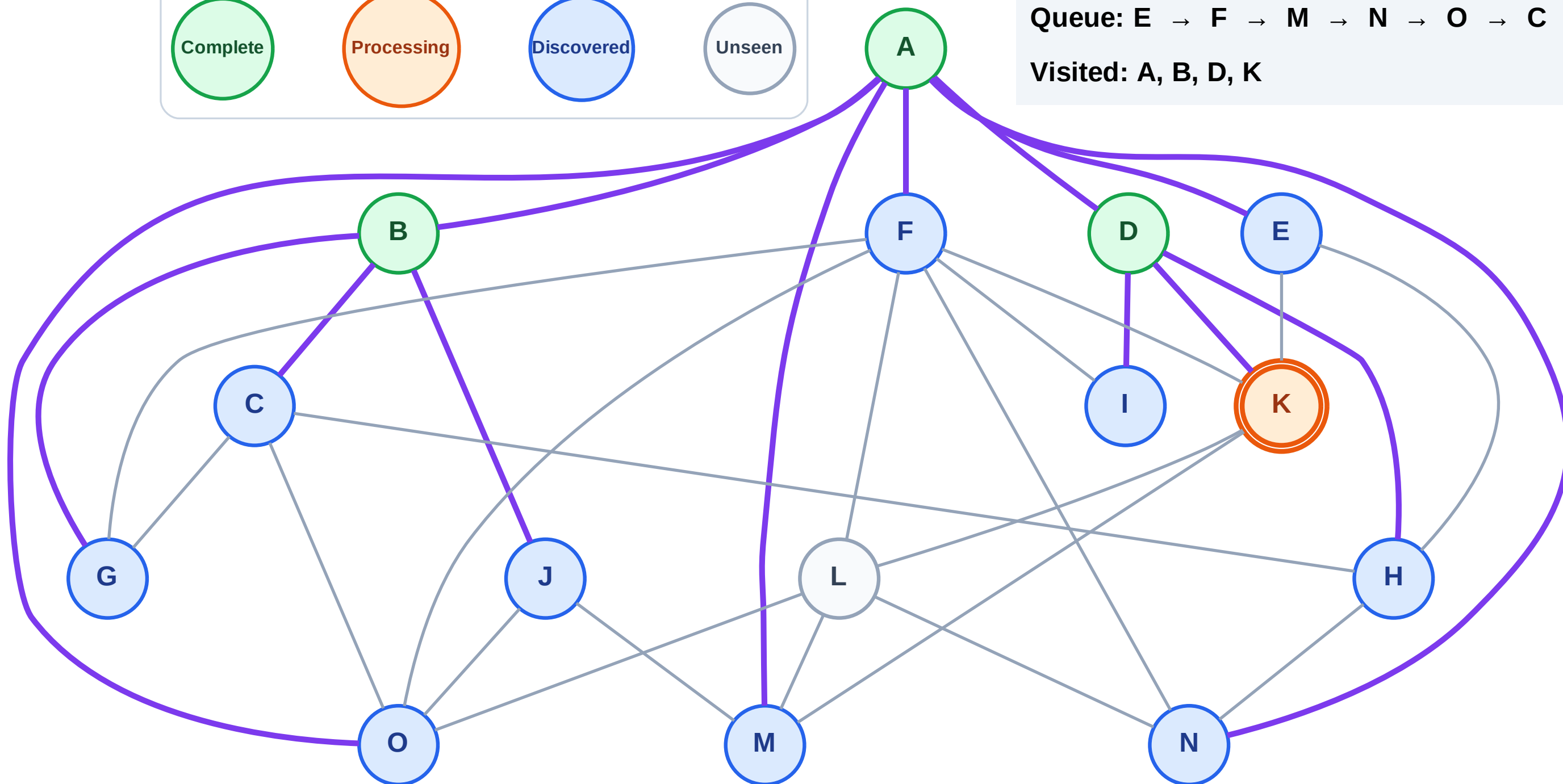
Discovered



Unseen

Queue: E → F → M → N → O → C → G → J → H → I

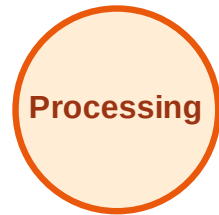
Visited: A, B, D, K



DFS Traversal

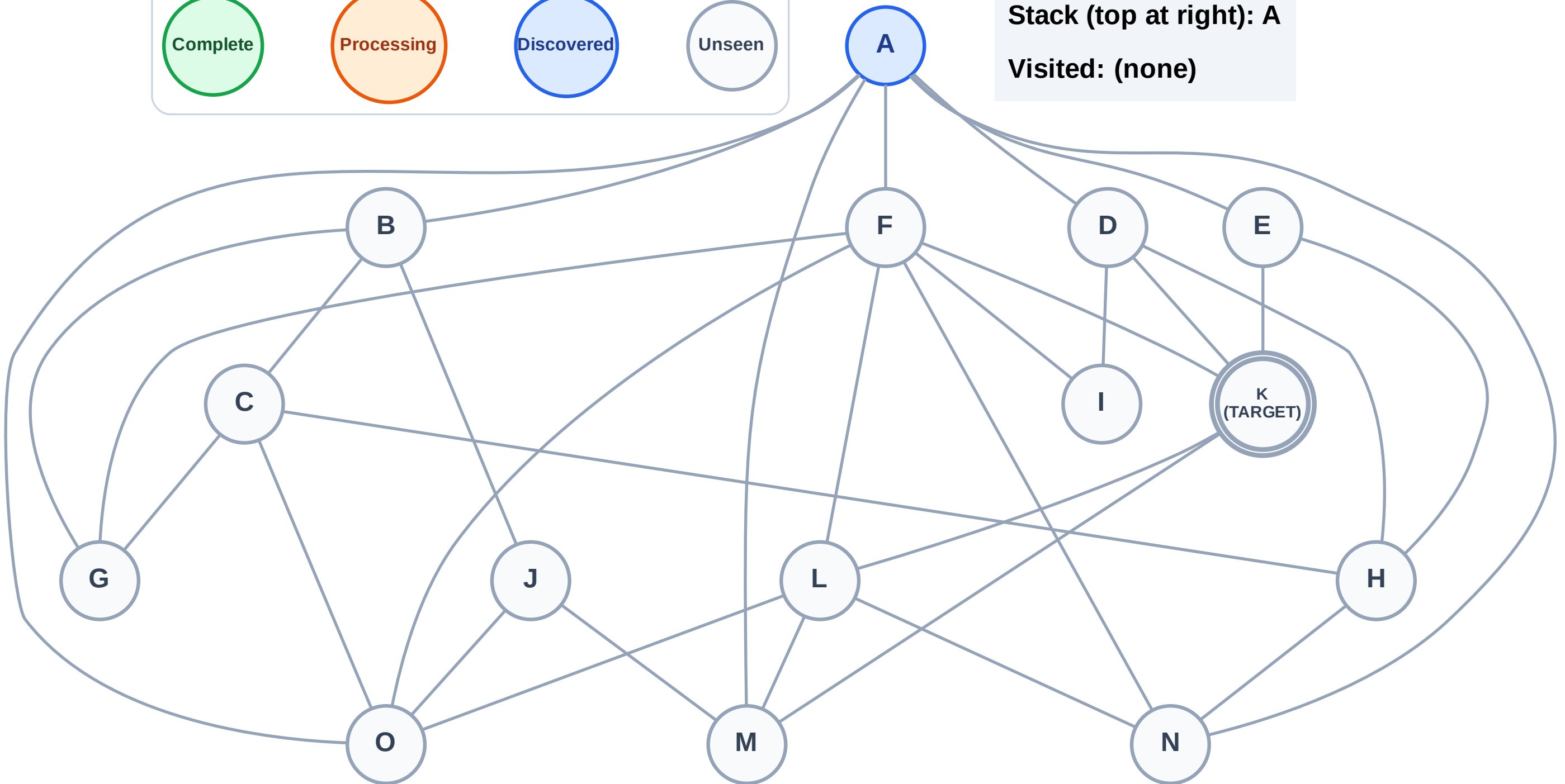
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

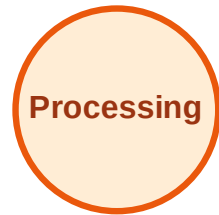
Visited: (none)



DFS Traversal

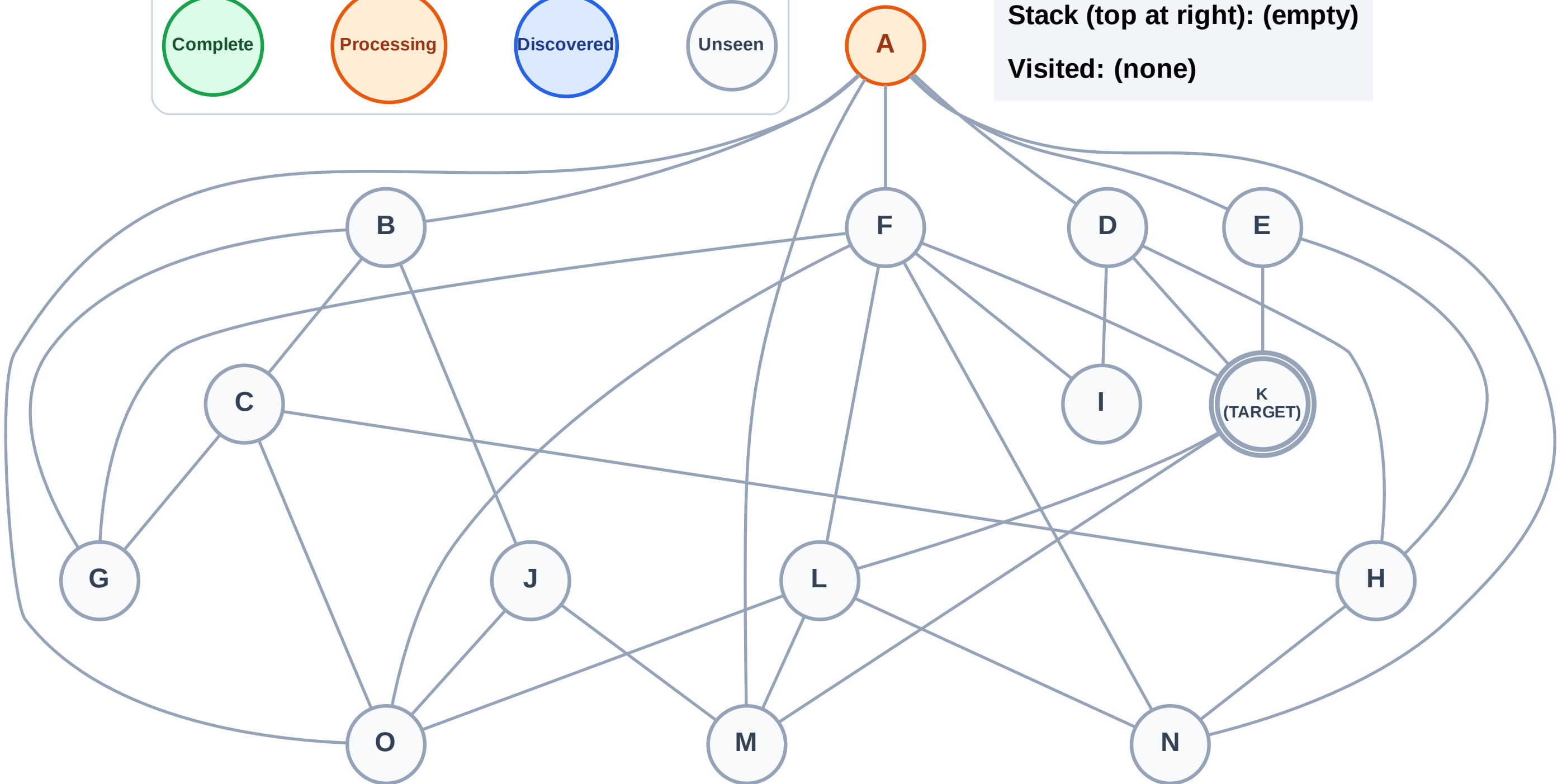
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

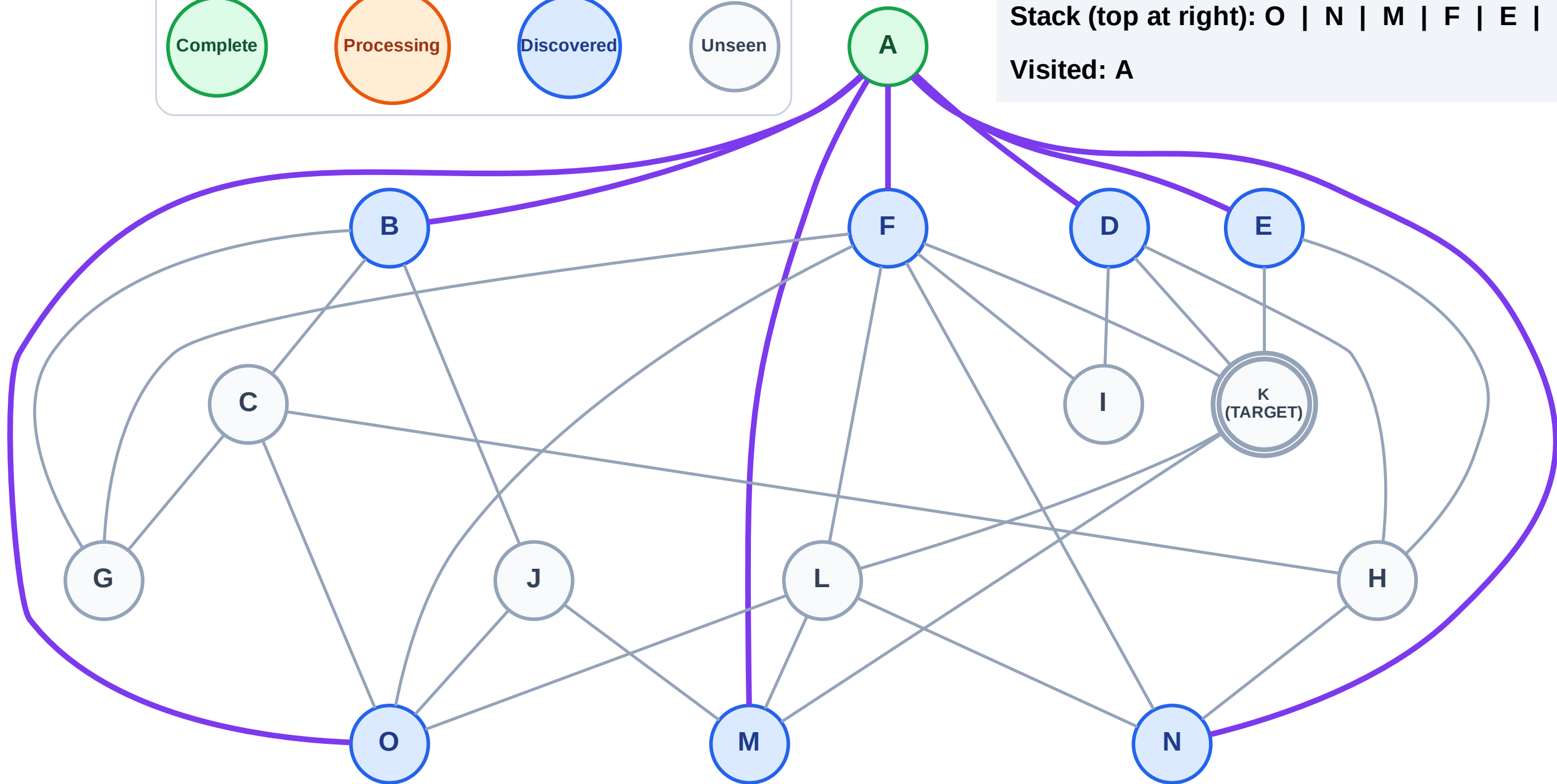
Step 3: Discover O, N, M, F, E, D, B from A

Legend



Stack (top at right): O | N | M | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

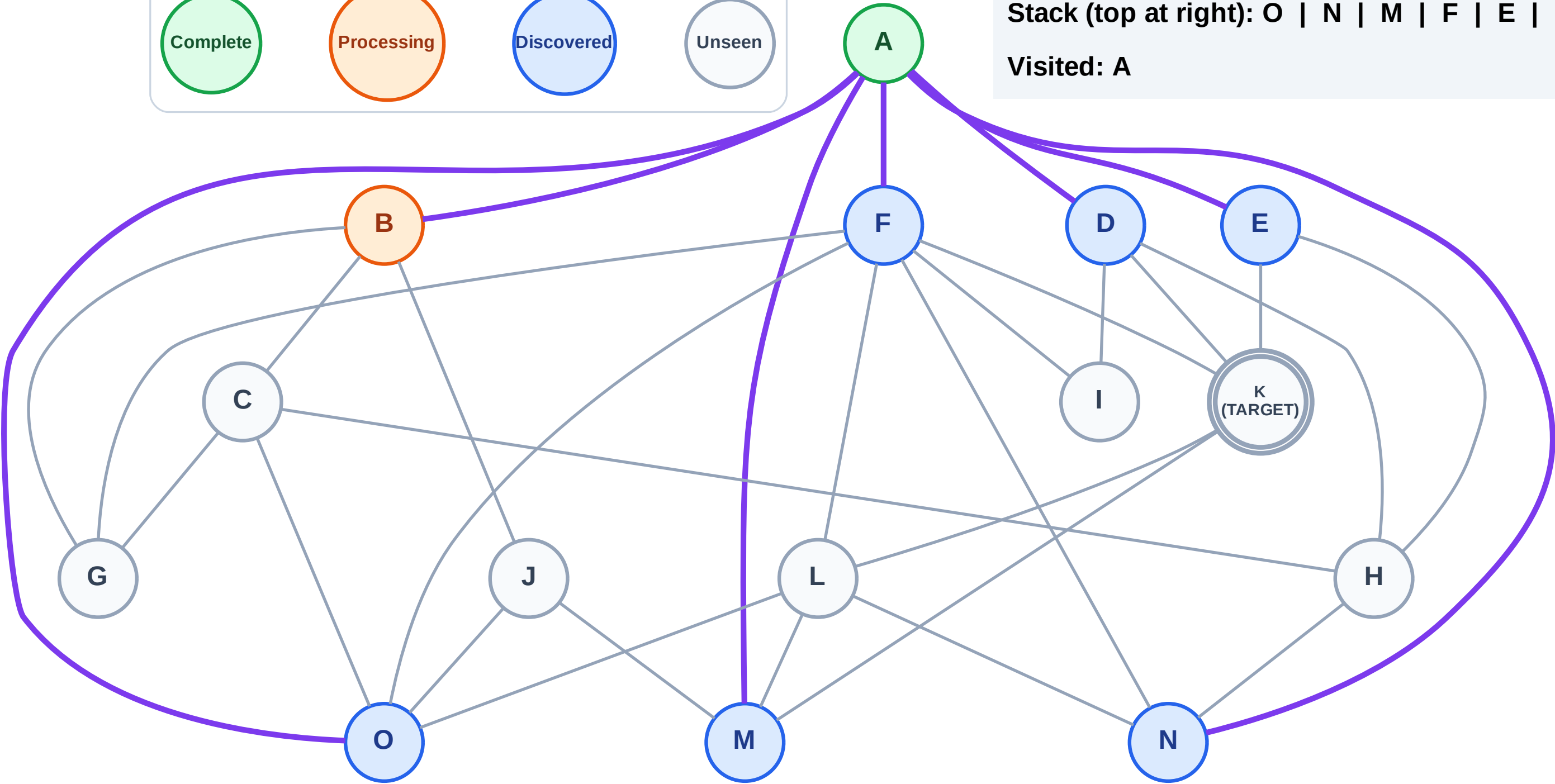
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D

Visited: A



DFS Traversal

Step 6: Process node C

Legend

Complete

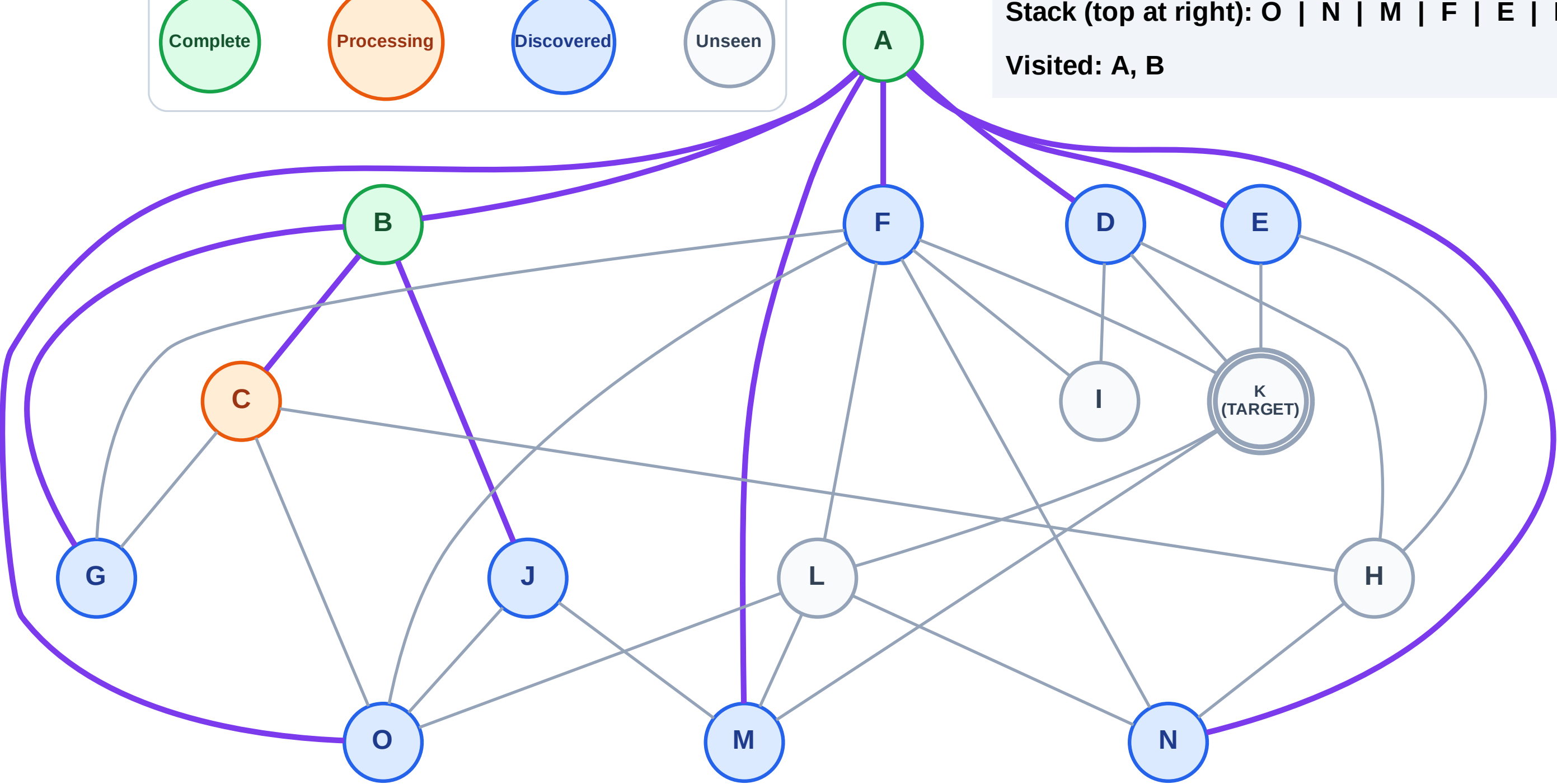
Processing

Discovered

Unseen

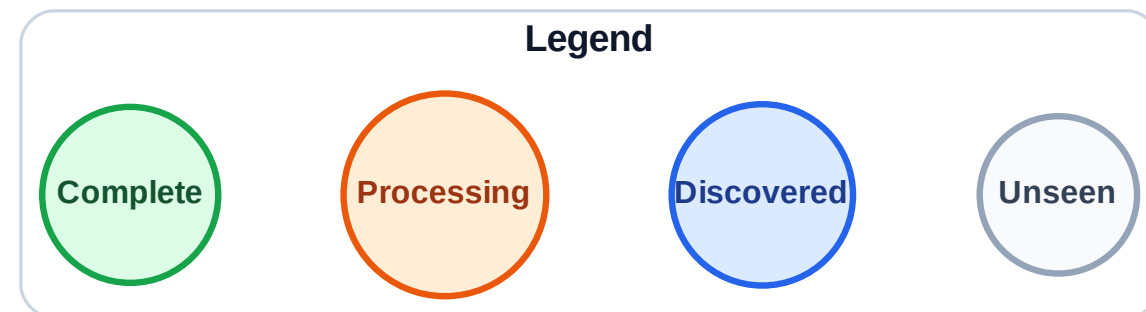
Stack (top at right): O | N | M | F | E | D | J | G

Visited: A, B



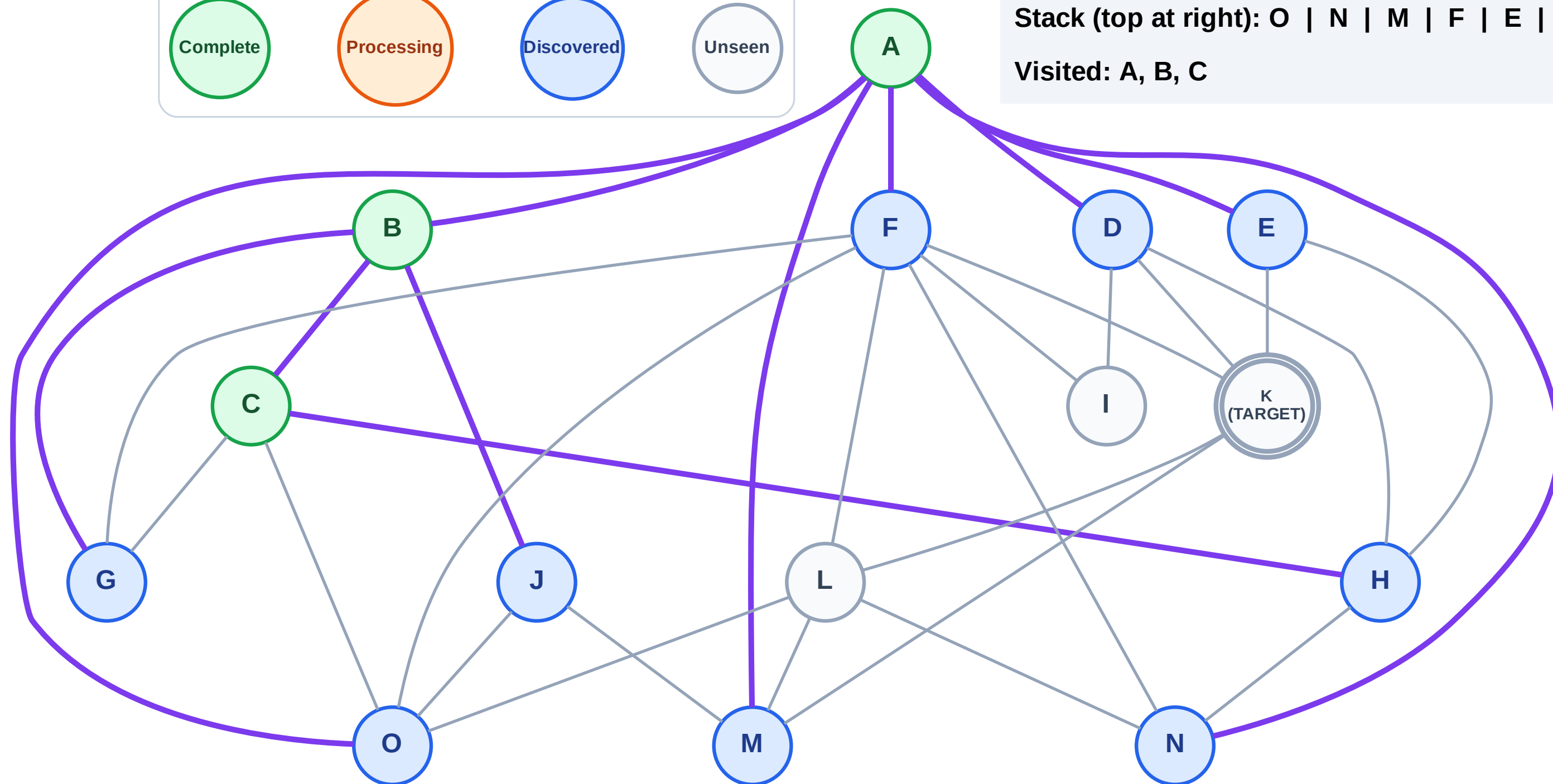
DFS Traversal

Step 7: Discover H from C



Stack (top at right): O | N | M | F | E | D | J | G | H

Visited: A, B, C



DFS Traversal

Step 8: Process node H

Legend

Complete

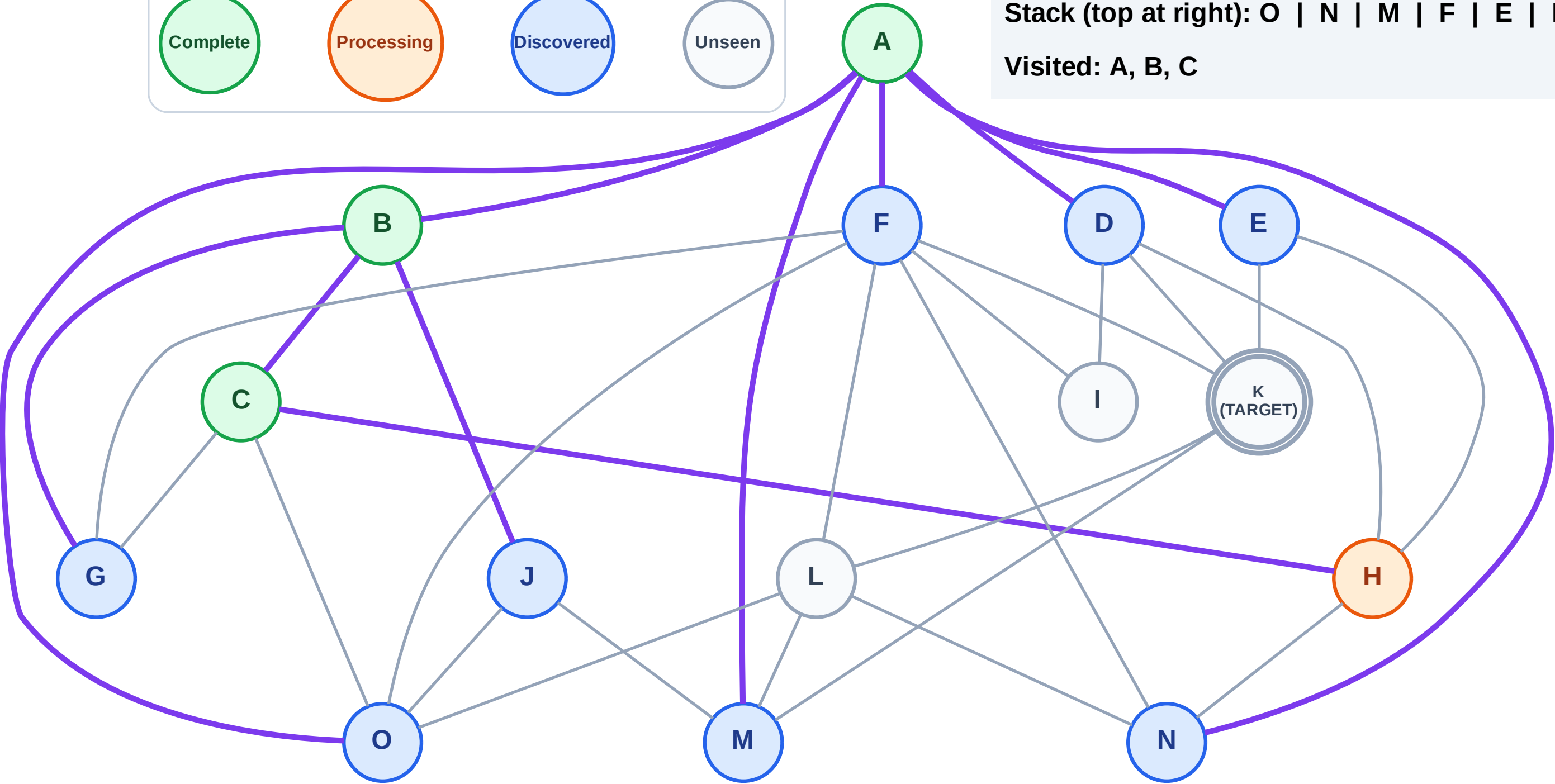
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D | J | G

Visited: A, B, C



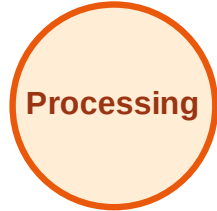
DFS Traversal

Step 9: No new neighbors from H

Legend



Complete



Processing



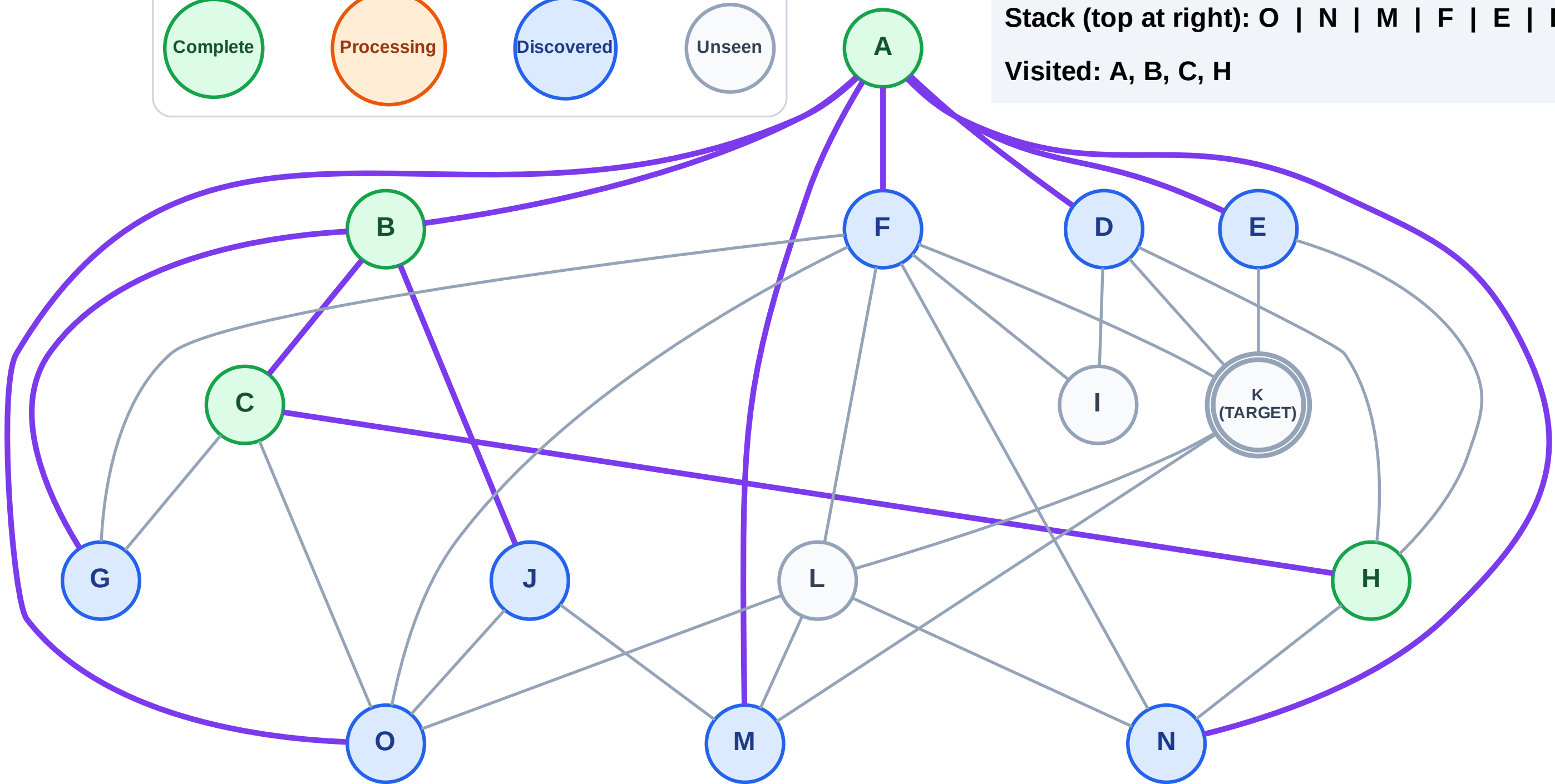
Discovered



Unseen

Stack (top at right): O | N | M | F | E | D | J | G

Visited: A, B, C, H



DFS Traversal

Step 10: Process node G

Legend

Complete

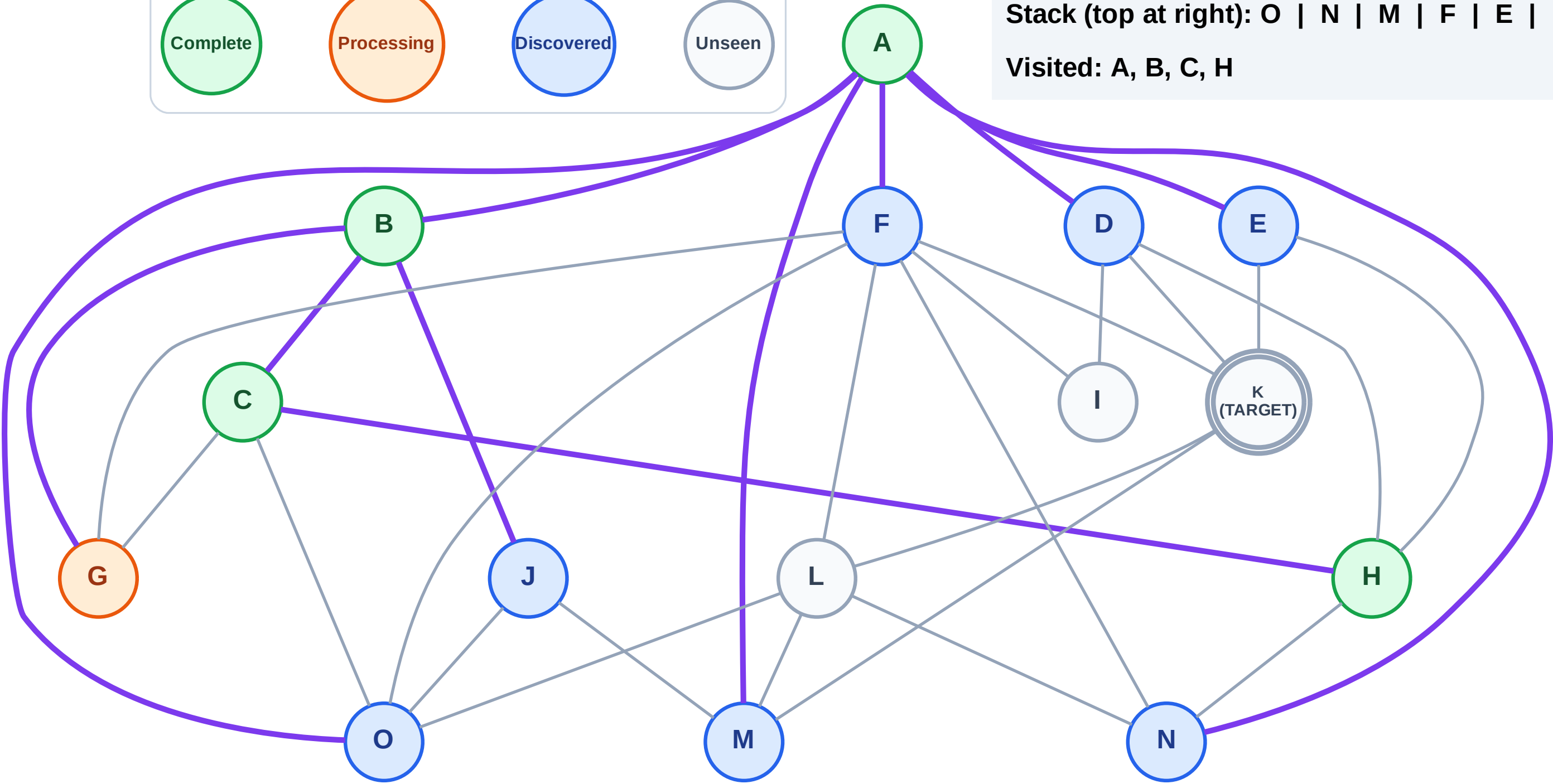
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D | J

Visited: A, B, C, H



DFS Traversal

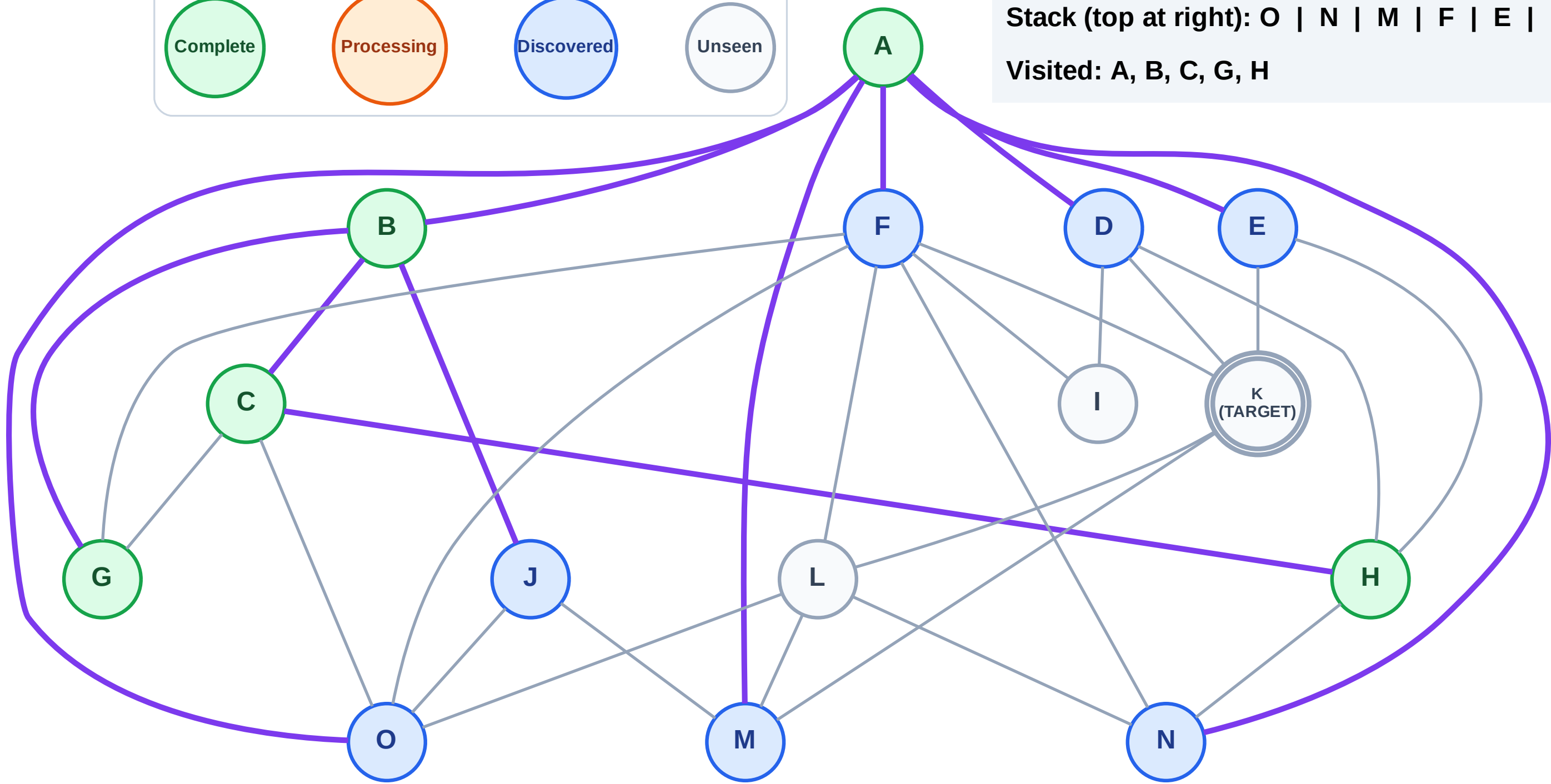
Step 11: No new neighbors from G

Legend



Stack (top at right): O | N | M | F | E | D | J

Visited: A, B, C, G, H



DFS Traversal

Step 12: Process node J

Legend

Complete

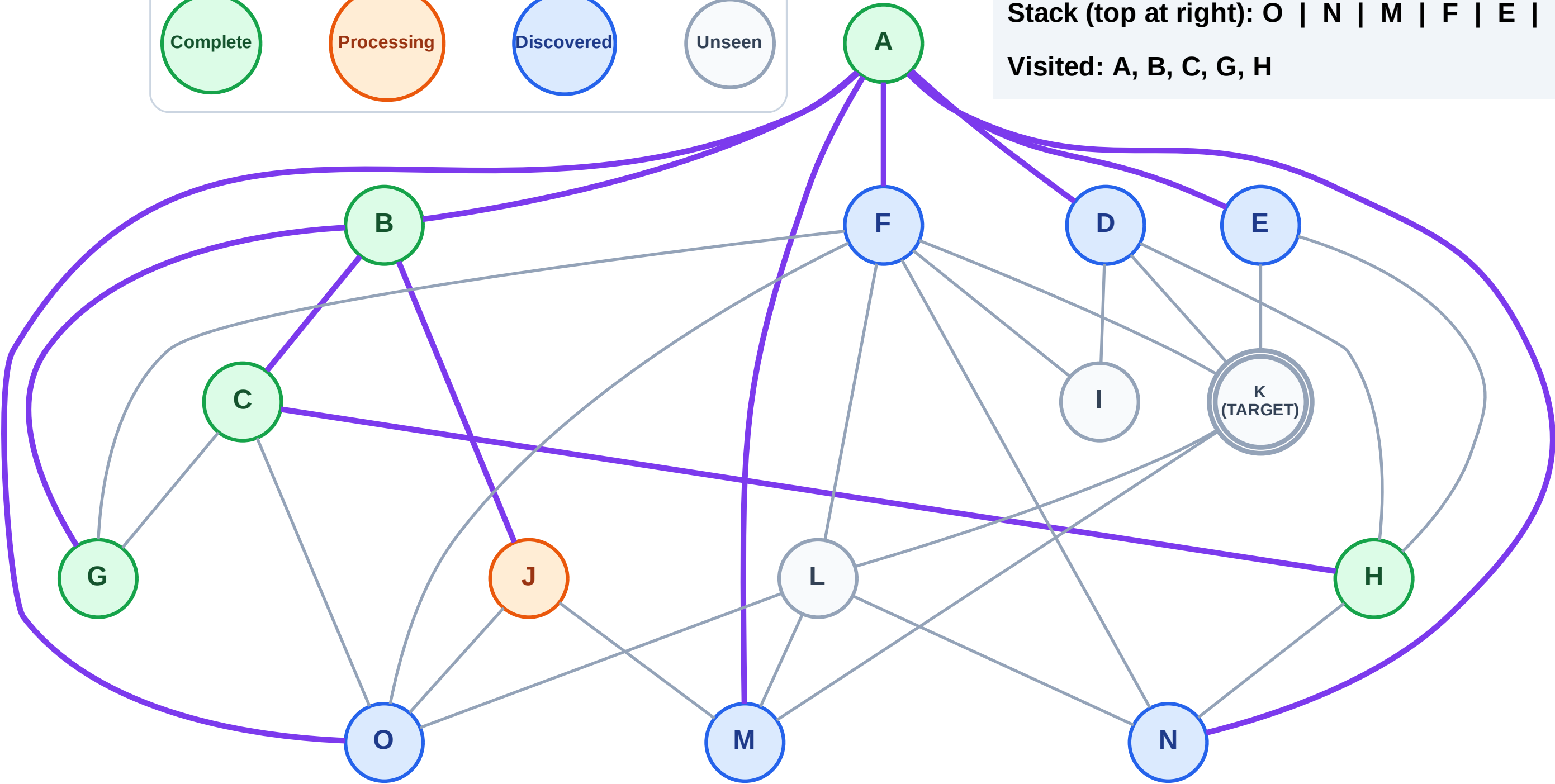
Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E | D

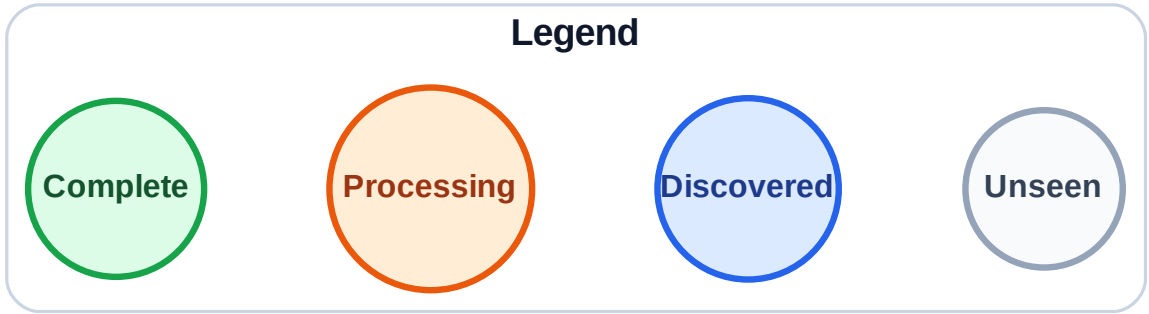
Visited: A, B, C, G, H



DFS Traversal

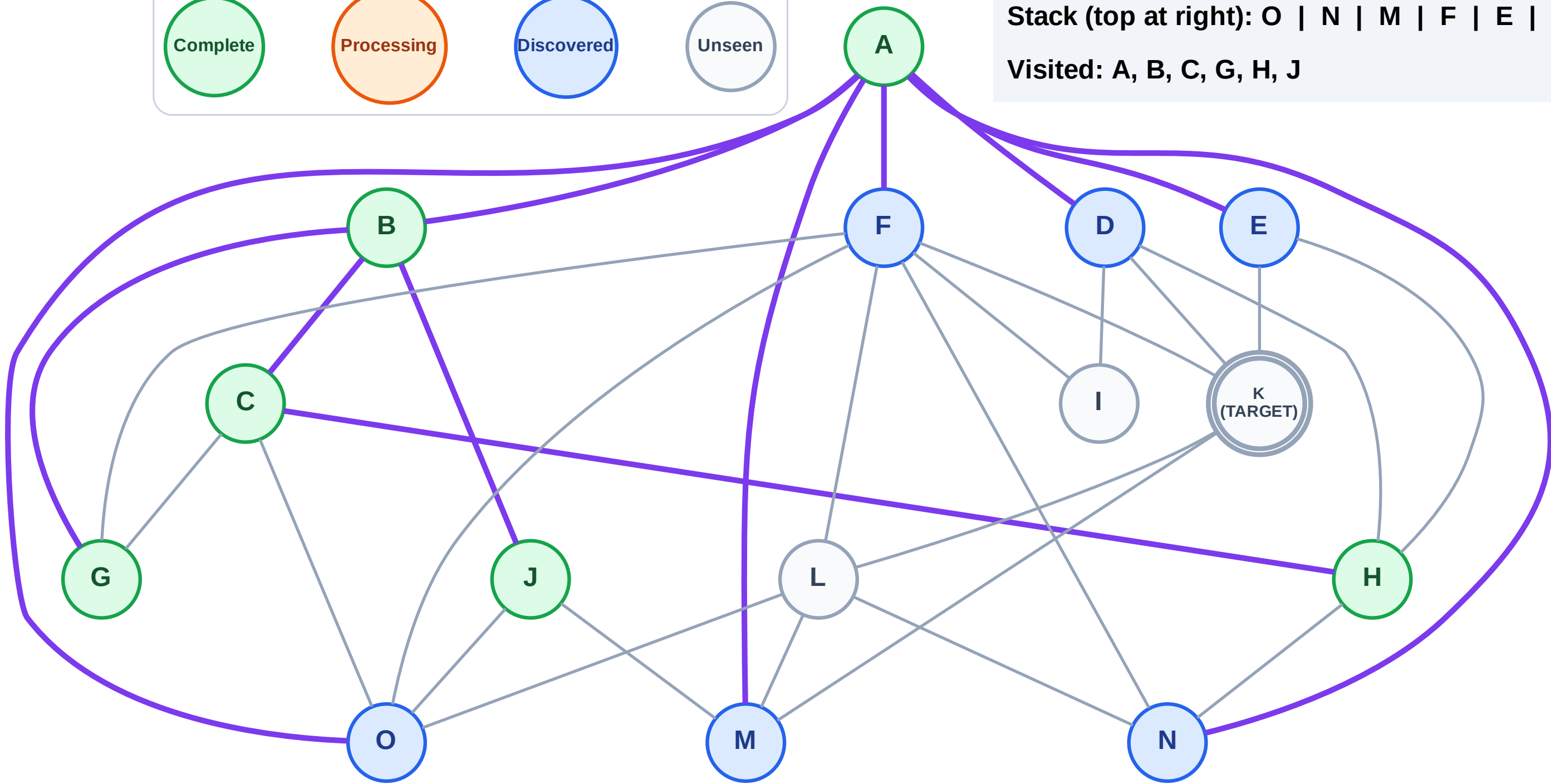
Step 13: No new neighbors from J

Legend



Stack (top at right): O | N | M | F | E | D

Visited: A, B, C, G, H, J



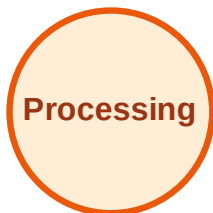
DFS Traversal

Step 14: Process node D

Legend



Complete



Processing



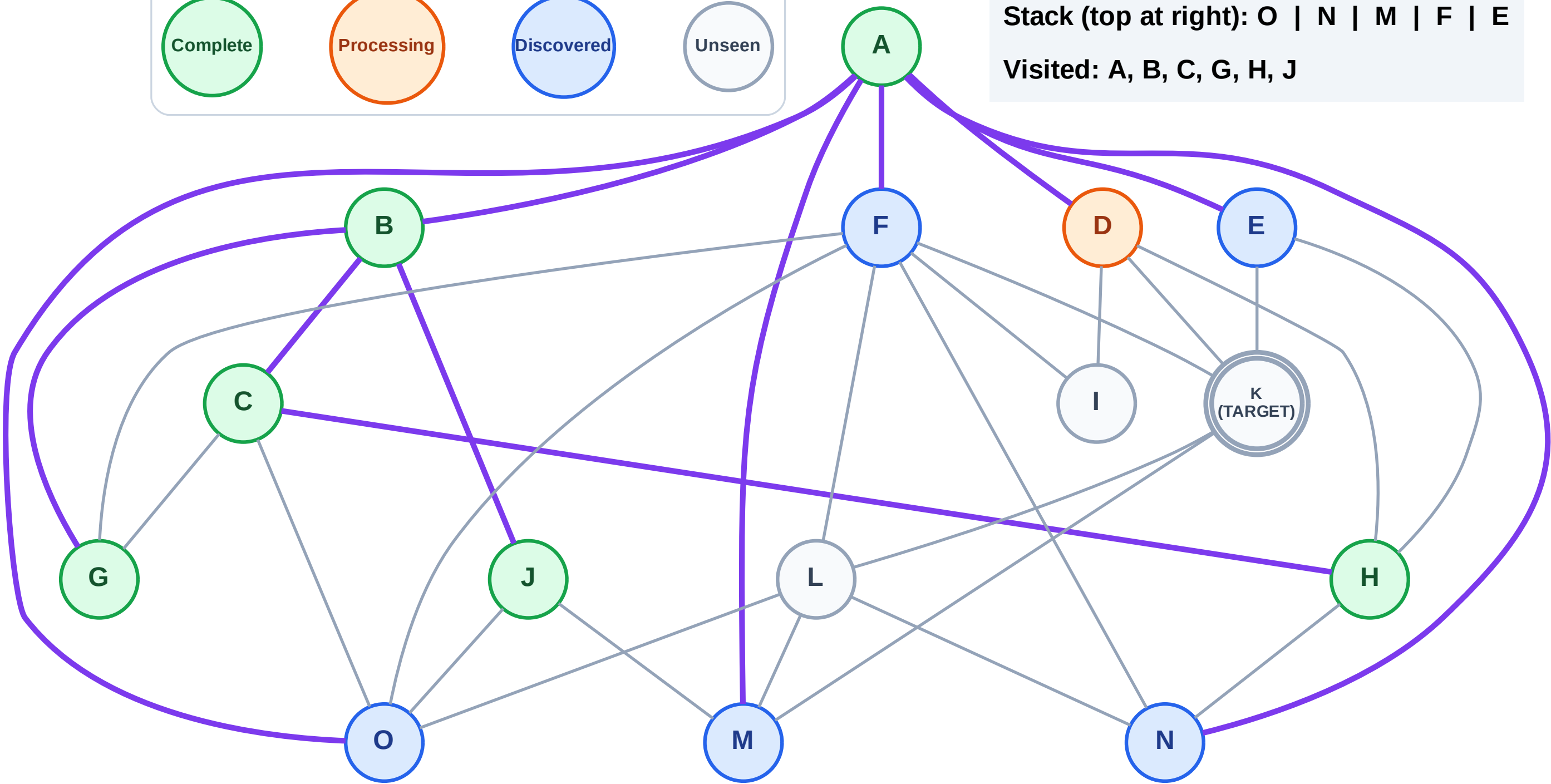
Discovered



Unseen

Stack (top at right): O | N | M | F | E

Visited: A, B, C, G, H, J



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M | F | E
Visited: A, B, C, D, G, H, J, K

